Quantum reservoir on NF-UNSW-NB15-v2: the full grid

14 October 2026 | replaces the report of 13 October | qrc_iontrap_sweep_alltoall.py with qrc_iontrap_core.py, as uploaded on 12 October, unchanged | trapped-ion reservoir, simulated with 512 shots

Bottom line

No reservoir beats the classical baseline. The grid is now complete: 6, 8, 10, 12, 14 and 16 qubits, each with 6, 8 and 10 time steps (L), 18 cells in all. In none is the reservoir ahead of the baseline by more than test-set chance, on either test.

Unseen attacks (zero-day): a bigger reservoir catches up, whether it grows in qubits or in time steps. Averaged over the three step counts, its AUC gap to the baseline goes from -0.020 at 6 qubits to -0.001 at 16. Averaged over the six sizes, it is -0.013 at L = 6, -0.013 at L = 8 and -0.005 at L = 10. At 14 and 16 qubits the average gap is -0.003 (interval -0.012 to +0.007): level with the baseline, not ahead of it.

Known attacks (supervised): behind or level; more time steps help, more qubits do not. The reservoir is behind in all 18 cells, by 0.001-0.024 AUC. The gap averages -0.013 at L = 6, -0.009 at L = 8 and -0.006 at L = 10, and shows no trend in qubit count.

What changed since the report of 13 October. 3 cells are new: L = 10 at 6, 8 and 10 qubits. All of them are level with the baseline. With them, the earlier statement that time steps make no consistent difference no longer holds. The 12 cells that were run again came out identical to 13 October, digit for digit.

How far to trust this. One data split and one random reservoir per qubit count. The intervals cover the limited test set only (sections 1 and 7).

Detection at 1% false alarms, the sweep's headline figure, cannot be judged on this test set. The 95% interval on the difference between the two arms is about ±6 percentage points (section 5).

1. What was run

Every qubit is coupled to every other; half the qubits take the data in and are measured and reset at each of L time steps, the other half carry memory. Each network flow is measured with **512 shots** and a single linear layer is trained on the measured averages. The **classical baseline** is the same linear layer trained directly on the input features, with no quantum step.

Qubits (inputs + memory)	L = 6	L = 8	L = 10
6 (3 + 3)	run again: identical	run again: identical	**new**
8 (4 + 4)	run again: identical	run again: identical	**new**
10 (5 + 5)	run again: identical	run again: identical	**new**
12 (6 + 6)	run again: identical	run again: identical	run again: identical
14 (7 + 7)	from 13 Oct	from 13 Oct	from 13 Oct
16 (8 + 8)	run again: identical	run again: identical	run again: identical
20 (10 + 10)	not run	not run	not run

On 14 October the sweep was run again for 6, 8, 10, 12 and 16 qubits with L = 6, 8 and 10, using the same two project scripts and the same settings as on 13 October (reservoir seed 1234, data split seed 42). 12 of those 15 cells had been run before. Every one reproduced the earlier run exactly: the same reservoir, the same simulated measurements and the same scores (compare_with_13oct.py). The other 3 are new. 14 qubits was not part of the rerun and is taken from 13 October.

Each cell uses 400 training flows, 300 test flows (150 benign, 150 attack) and 150 flows of one attack family, **Exploits**, hidden from training. **Supervised** means telling benign from attack families seen in training. **Zero-day** means a detector trained on benign traffic only must flag the hidden family. Scores are AUC (1.0 perfect, 0.5 chance).

How differences are judged. Every difference between the reservoir and the baseline is given with a 95% interval. It is obtained by drawing the test flows again at random 4,000 times and scoring both arms on the same draw each time (check_grid_effects.py, which first reproduces every AUC of the sweep from the saved features). A cell is called **behind** when its whole interval lies below zero and **level** when the interval includes zero. The interval reflects the limited test set only. The training flows, the reservoir's random couplings and the shot sampling are held fixed, so the real uncertainty is larger.

2. The dataset

	v1 (used until 12 Oct)	v2 (this report)
Features per flow in the file	10	43
Flows in the file	1.48 million	2.39 million
Features used	10, ports included	37: IP addresses, ports and TTLs dropped
Flows used	1.48 million	241,165 after removing repeated rows
Inputs after encoding	24	53
Attack share of flows used	4.3%	17.2%

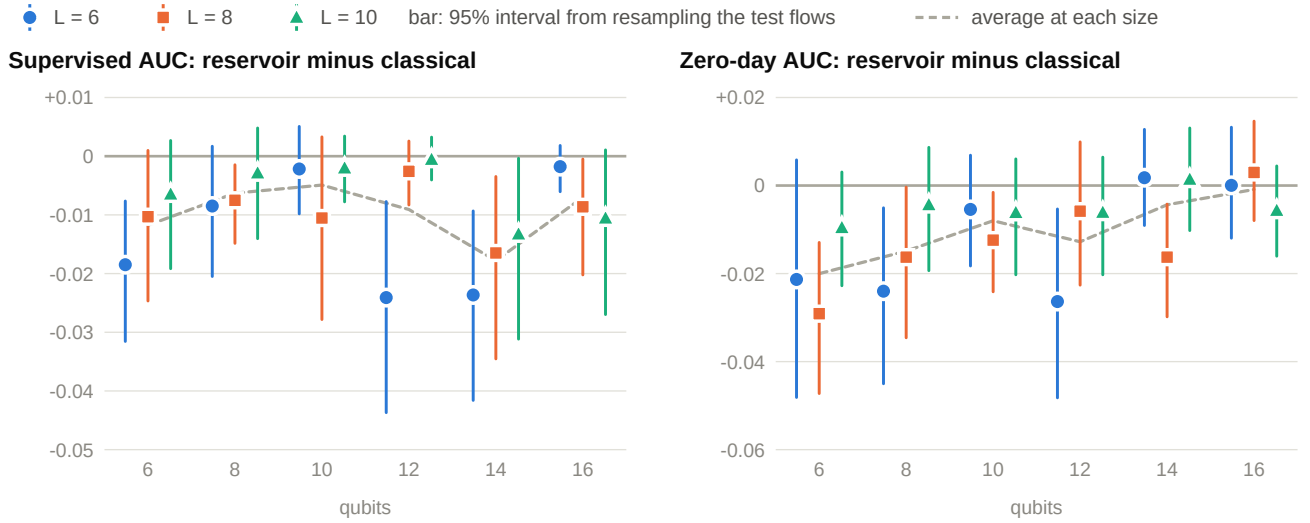
v2 is the same captured traffic with the same labels, exported with far more detail per flow. The preparation follows the project's own NF-UNSW-NB15-v2.npz found in the archive: 37 features, log1p applied to every one, and rows that repeat an earlier row removed (90% of the file once the identifying columns are dropped). TCP flags are one-hot encoded as the core does for v1, giving 53 inputs. A reservoir with n input qubits and L steps has 2 x n x L encoding slots. 3 small cells have fewer slots than inputs: 6 qubits (L = 6 and 8) and 8 qubits (L = 6). They are given the leading 36 or 48 principal components, which hold 99.88% of the variance or more, and their classical baseline gets the same components. Every other cell takes in all 53 inputs and fills its spare slots by reading inputs again (up to 3.0 times over at the largest cell).

3. Results

		Supervised AUC (known attacks)				Zero-day AUC (unseen attacks)			
Qubits	L	Quantum	Classical	Diff.	95% interval	Quantum	Classical	Diff.	95% interval
6	6	0.965	0.984	**-0.018**	**-0.032 to -0.008**	0.867	0.888	-0.021	-0.048 to +0.006
6	8	0.975	0.985	-0.010	-0.025 to +0.001	0.859	0.888	**-0.029**	**-0.047 to -0.013**
6	10*	0.980	0.986	-0.006	-0.019 to +0.003	0.878	0.888	-0.010	-0.023 to +0.003
8	6	0.977	0.985	-0.008	-0.020 to +0.002	0.864	0.888	**-0.024**	**-0.045 to -0.005**
8	8	0.979	0.986	**-0.008**	**-0.015 to -0.001**	0.871	0.888	-0.016	-0.035 to 0.000
8	10*	0.984	0.986	-0.003	-0.014 to +0.005	0.883	0.888	-0.004	-0.019 to +0.009
10	6	0.984	0.986	-0.002	-0.010 to +0.005	0.882	0.888	-0.005	-0.018 to +0.007
10	8	0.976	0.986	-0.011	-0.028 to +0.003	0.875	0.888	**-0.012**	**-0.024 to -0.002**
10	10*	0.984	0.986	-0.002	-0.008 to +0.003	0.882	0.888	-0.006	-0.020 to +0.006
12	6	0.962	0.986	**-0.024**	**-0.044 to -0.008**	0.861	0.888	**-0.026**	**-0.048 to -0.005**
12	8	0.984	0.986	-0.003	-0.008 to +0.003	0.882	0.888	-0.006	-0.023 to +0.010
12	10	0.986	0.986	-0.001	-0.004 to +0.003	0.882	0.888	-0.006	-0.020 to +0.006
14	6	0.963	0.986	**-0.024**	**-0.042 to -0.009**	0.889	0.888	+0.002	-0.009 to +0.013
14	8	0.970	0.986	**-0.016**	**-0.035 to -0.003**	0.871	0.888	**-0.016**	**-0.030 to -0.004**
14	10	0.973	0.986	-0.013	-0.031 to 0.000	0.889	0.888	+0.001	-0.010 to +0.013
16	6	0.985	0.986	-0.002	-0.006 to +0.002	0.888	0.888	0.000	-0.012 to +0.013
16	8	0.978	0.986	-0.009	-0.020 to 0.000	0.891	0.888	+0.003	-0.008 to +0.015
16	10	0.976	0.986	-0.011	-0.027 to +0.001	0.882	0.888	-0.006	-0.016 to +0.004

* new on 14 October. Diff. = quantum minus classical; negative means the reservoir is behind. The 95% interval is described in section 1; **bold** marks a difference whose interval lies wholly below zero. The quantum columns use only the measured features that rise above their own shot noise (21 of 24 at the smallest cell, 80 of 96 at the largest); using all of them moves either AUC by at most 0.005.

How the gap changes with reservoir size



Each point is one cell of the table above: the reservoir's AUC minus the classical baseline's. Zero means level with the baseline; below zero means behind.

4. Reading the results

The same differences, laid out as the grid they come from. Reading across a row shows the effect of time steps; reading down a column shows the effect of qubits.

	Zero-day gap				Supervised gap			
Qubits	**L = 6**	**L = 8**	**L = 10**	**Average**	**L = 6**	**L = 8**	**L = 10**	**Average**
6	-0.021	**-0.029**	-0.010	**-0.020**	**-0.018**	-0.010	-0.006	**-0.012**
8	**-0.024**	-0.016	-0.004	**-0.015**	-0.008	**-0.008**	-0.003	-0.006
10	-0.005	**-0.012**	-0.006	-0.008	-0.002	-0.011	-0.002	-0.005
12	**-0.026**	-0.006	-0.006	-0.013	**-0.024**	-0.003	-0.001	**-0.009**
14	+0.002	**-0.016**	+0.001	-0.004	**-0.024**	**-0.016**	-0.013	**-0.018**
16	0.000	+0.003	-0.006	-0.001	-0.002	-0.009	-0.011	-0.007
Average	**-0.013**	**-0.013**	-0.005	-0.010	**-0.013**	**-0.009**	-0.006	-0.009

Gap = reservoir minus classical AUC. **Bold** = behind the baseline by more than test-set chance (interval wholly below zero); this is judged for the row and column averages as well.

- **Unseen attacks, cell by cell.** The reservoir is behind the baseline by more than test-set chance in 5 cells: 6 qubits (L = 8), 8 qubits (L = 6), 10 qubits (L = 8), 12 qubits (L = 6) and 14 qubits (L = 8). In the other 13 it is level. One of them sits on the line, with an interval that ends at 0.000: 8 qubits (L = 8). 3 of the level cells are nominally ahead, by at most +0.003, which is well inside the interval: 14 qubits (L = 6 and 10) and 16 qubits (L = 8).
- **Unseen attacks and qubits.** The gap narrows as qubits are added: -0.017 on average at 6 and 8 qubits, -0.003 at 14 and 16, a difference of +0.015 (interval +0.004 to +0.027). Fitted across the grid, that is +0.0017 AUC per added qubit (interval +0.0005 to +0.0031).
- **Unseen attacks and time steps.** L = 10 is better than L = 8 by +0.008 (interval +0.003 to +0.013). L = 8 is no better than L = 6: the difference is 0.000 (interval -0.006 to +0.006). L = 10 against L = 6 is +0.007, with an interval that starts at 0.000 (0.000 to +0.015). So steps help on this test, but modestly. The effect is clearest at the small sizes: at 6 and 8 qubits the gap at L = 10 is -0.010 and -0.004, against 0.016-0.029 behind at L = 6 and 8.
- **Much of this is capacity.** The 3 cells with fewer encoding slots than inputs (section 2) average -0.025 on unseen attacks; the 15 cells that take in every input average -0.007, a difference of +0.018 (interval +0.004 to +0.032). 6 qubits at L = 10 is the smallest reservoir that takes in all 53 inputs, and it is level with the baseline.

Among the 15 cells that take in every input the remaining effects are smaller: +0.0012 per added qubit (interval +0.0001 to +0.0023) and +0.0013 per added step (interval -0.0001 to +0.0028).

- **Known attacks.** The reservoir is behind in all 18 cells. In 5 the gap is more than test-set chance: 6 qubits (L = 6), 8 qubits (L = 8), 12 qubits (L = 6) and 14 qubits (L = 6 and 8). In the other 13 it is level. 2 of them sit on the line, with an interval that ends at 0.000: 14 qubits (L = 10) and 16 qubits (L = 8). More time steps help: L = 10 is better than L = 6 by +0.007 (interval +0.001 to +0.013). More qubits do not: the fitted change is -0.0002 AUC per added qubit (interval -0.0013 to +0.0006), and at 14 and 16 qubits the average gap is still -0.012 (interval -0.025 to -0.003).
- **Qubit count and chance are mixed together; step count is not.** Each qubit count uses one random draw of the reservoir's couplings (seed 1234). A difference between two qubit counts is therefore partly the effect of size and partly the luck of the draw, and only more draws can separate them. The three step counts at one qubit count share the same couplings, so a difference between L = 6, 8 and 10 does not have this problem.

5. Detection at a fixed false-alarm rate

The sweep's closing note calls the share of hidden attacks caught at 1% false alarms "the pitch number". These are the figures, with the share at 5% as well:

Qubits	6			8			10			12			14			16		
L (time steps)	**6**	**8**	**10**	**6**	**8**	**10**	**6**	**8**	**10**	**6**	**8**	**10**	**6**	**8**	**10**	**6**	**8**	**10**
Reservoir, caught at 1%	17%	1%	5%	11%	5%	2%	6%	3%	5%	6%	3%	5%	6%	5%	7%	5%	5%	6%
Classical, caught at 1%	6%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%	9%
Reservoir, caught at 5%	31%	7%	24%	32%	16%	17%	28%	8%	10%	17%	23%	25%	8%	13%	21%	15%	13%	15%
Classical, caught at 5%	8%	11%	11%	11%	11%	11%	11%	11%	11%	11%	11%	11%	11%	11%	11%	11%	11%	11%

Read at face value, the reservoir catches fewer hidden attacks than the baseline at 1% in 16 of the 18 cells and more at 5% in 14. Neither reading can be relied on, for two reasons.

- **The test set is too small for these figures.** With 150 benign test flows, the 1% threshold falls between the second and third highest-scoring benign flows and the 5% threshold between the eighth and ninth. Resampling the test flows as in section 1 gives a 95% interval on the reservoir-minus-classical difference of typically ±6 percentage points at 1% and ±26 at 5%. The interval includes zero in 17 of 18 cells at 1% and in all 18 at 5%.
- **They swing with a detector setting that barely moves the AUC.** The classical detector, rebuilt outside the sweep (check_classical_baseline_v2.py, 13 October) and told to use different numbers of directions of its benign-traffic model, gives:

Directions used by the classical detector	Zero-day AUC	Caught at 1%	Caught at 5%
32	0.872	9%	36%
36	0.875	2%	7%
40	0.888	1%	8%
41 (this run)	0.888	9%	11%
53 (all of them)	0.890	9%	11%

Across these settings the AUC moves by 0.018, while the share caught at 5% moves between 7% and 36% and at 1% between 1% and 9%. The reservoir shows the same disconnect: the cell with the highest zero-day AUC (16 qubits, L = 8: 0.891) catches 13% at 5%, while the one that catches most (8 qubits, L = 6: 32%) has an AUC of 0.864. AUC uses every test flow rather than the top few, which is why sections 3 and 4 rest on it. A usable figure at 1% false alarms needs a benign test set in the thousands.

6. What this means

- On unseen attacks, the test the project is built around, what matters most is whether the reservoir is big enough to take the whole flow in. The cells that are not are behind a plain classical detector by 0.025 on average. The cells that are, by having more qubits or more time steps, are level with one in 12 of 15 cases. Matching is not adding: no cell is ahead, and the best single result (+0.003) is far inside its interval.
- The sweep's closing note asks which axis is worth the gate budget. Time steps help on known attacks and, more modestly, on unseen attacks; qubits help on unseen attacks only. Time steps are also the cheaper route to the same score. 8 qubits at L = 10 is level with the baseline on both tests with 308 two-qubit gates per flow; 16 qubits at L = 6 is level on both with 840. The core's rough estimate of an error-free run is 77% for the first and 50% for the second.
- Moving to v2 did more than any change to the reservoir. It raised the classical zero-day AUC by 0.060 and the reservoir's by 0.070-0.103 at the six cells run on both (6, 8, 10 qubits at L = 6 and 8). The reservoir's zero-day deficit in those cells was 0.039-0.066 on v1 and is 0.005-0.029 on v2. On known attacks the two arms were level on v1 (differences of -0.003 to +0.009) and the reservoir is slightly behind on v2 (-0.018 to -0.002).
- The open question is whether a still bigger reservoir moves past the baseline or stays level with it. These results cannot say: at 14 and 16 qubits the interval on the average zero-day gap runs from -0.012 to +0.007. Two ways to test it: 20 qubits, about 50 hours on this machine; or more time steps at 8 to 12 qubits, for example L = 12 and 14, where a cell at L = 10 took at most 4 minutes. These are suggestions, not something this run tested.

7. What this run does not settle

- **One data split.** Every cell is compared with the same baseline on the same 300 test flows and trained on the same 400 flows. The intervals in this report cover the choice of test flows only. The rerun used the same split as before (seed 42), so it confirms that the computation is reproducible, not that the result would survive a different split. The sweep has a split-seed option for that; it was not used.
- **One reservoir per qubit count.** See the last point of section 4. Several random draws per size would be needed to be sure the effect of qubits on unseen attacks belongs to size and not to the particular draws.
- **Shot noise or the model?** The script offers an "exact" mode that removes shot noise and gives the ceiling no number of shots could exceed. It was left off in the uploaded file and was not run. If the ceiling is also level with the baseline, more shots cannot help.
- **Simulation only.** The results include shot sampling but no gate or measurement errors. The core's own rough estimate of the chance that one circuit runs with no error at all, from the gate fidelities written into it, falls from 91% (6 qubits, L = 6) to 34% (16 qubits, L = 10). Bigger reservoirs score better here and would suffer more on hardware.
- **A modest baseline.** The classical arm is a linear model and a simple distance detector. A tuned tree ensemble, the usual strong baseline for this kind of data, was not run.
- **One hidden attack family.** The zero-day test hides Exploits only. Other families may be easier or harder to flag.
- **v1 against v2.** The two datasets differ in features, in the removal of repeated rows, in the attack share and in the log scaling, so the before/after figures show that both arms improved, not which change caused it.

8. Settings, computing time and files

backend = numpy (exact statevector with sampled shots) | shots = 512 | coupling normalisation = sk | dt = 0.3 | input gain = 3.0 | final basis = z | reservoir seed = 1234 | split seed = 42 | correlators = False | sampling = balanced | padding = reupload | hidden family = Exploits

Simulation time per cell on this machine (Apple M5 Ultra, 24 processes): 1 to 8 seconds at 6 and 8 qubits, 32 to 47 seconds at 10, 2 to 4 minutes at 12, 8 to 11 minutes at 14 (13 October) and 45 to 70 minutes at 16. 20 qubits was timed on single circuits at about 13, 17 and 20 hours for L = 6, 8 and 10.

Folder of the rerun: Downloads/qrc_run_2026-10-14_v2data_rerun/. It holds the two project scripts as uploaded; the wrappers and parallel_numpy.py, identical to 13 October; the logs; the results in results_large_6-8-10-12q/ and results_large_16q/ (sweep_summary.json and .csv, plus a results, features and reservoir file per cell); compare_with_13oct.py with rerun_comparison.json; check_grid_effects.py with grid_effects_check.json; and make_report_full_grid.py, which produced this document. The 14-qubit cells and the classical-detector check are in Downloads/qrc_run_2026-10-13_v2data/.